

Two-round subsampled randomized Hadamard transforms: canonical prescriptions, whole-sketch Nyström energy, and adaptive frames

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Abstract

Two rounds of sign randomization and unitary mixing admit smaller certified subspace embeddings when the sampling density is retained throughout the moment argument. Starting from a previously established row-factor estimate Λ_q , we prove the explicit prescription

$$q = \left\lceil \log_4 \frac{6d}{5\delta} \right\rceil, \quad M = \min\{n, \lceil 20\Lambda_q/\varepsilon^2 \rceil\}.$$

It reduces the earlier sufficient coefficient 72 to 20, without changing the sketch distribution; direct evaluation of the resulting moment bound can reduce the certified width further. Independently of the imported operator estimates, we derive exact fourth-moment identities for one without-replacement sketch and the Nyström bound $\mathbb{E}\|K - \mathcal{N}_K(S^*)\|_F^2 \leq 2(1 - M/n)(\text{tr } K)^2/M$ for real flat transforms. These identities support nonadaptive trace and diagonal estimators. We also transfer fixed-frame guarantees to a restricted form of sketch-dependent frame selection using the classical comparison between adaptive and volume sampling.

Contents

1	Introduction and main results	2
1.1	The model	2
1.2	Main results	3
1.3	Contributions and relation to earlier work	4
1.4	Roadmap	5
2	Imported results and limited independence	6
3	The exact moment root	7
3.1	From Bernoulli to fixed-size sampling	7
3.2	The root	8
4	The canonical prescription	9
4.1	The chain of constants	10

5	The direct deterministic selector	11
6	Adaptive frames selected by randomly pivoted Cholesky	13
7	Whole-sketch Nyström energy	15
7.1	Trace and diagonal estimation	18
8	Fresh-row drift	19
9	Sketched randomly pivoted Cholesky	21
10	Numerical evaluation	22
11	Open problems	25

1 Introduction and main results

A structured embedding is useful only when its row budget is explicit. For the two-round transform studied here, previous operator estimates already give a dimension term plus a quadratic confidence term. This paper asks a narrower quantitative question: how much of the stated row budget comes from the operator estimate, and how much comes from the final scalar calibration?

The main answer is that retaining the actual sampling density and the exact Bernoulli-to-fixed-size comparison reduces one sufficient coefficient from 72 to 20. The sketch and the imported row-factor estimate are unchanged. A separate fourth-moment calculation describes what one entire sketch can do for Nyström approximation. Keeping these two arguments separate makes their assumptions and their respective strengths explicit.

1.1 The model

Let $A, B \in \mathbb{C}^{n \times n}$ be unitary, with coherences

$$\alpha = n \max_{j,a} |A_{ja}|^2, \quad \beta = n \max_{a,i} |B_{ai}|^2. \quad (1.1)$$

Both are at least one, and both equal one exactly when the matrix is *flat*, $|A_{ja}|^2 = 1/n$; the normalized Walsh–Hadamard matrix H_n (n a power of two) and the normalized discrete Fourier matrix are flat. Let D_x, D_y be diagonal matrices of unbiased signs, the two families being independent, and let $\mathcal{T} \subseteq [n]$ be a uniform M -element subset, independent of the signs. With $R_{\mathcal{T}}$ the coordinate restriction to $\mathcal{T}$, the *two-round sketch* is

$$Q = AD_yBD_x, \quad S = \sqrt{n/M} R_{\mathcal{T}}Q \in \mathbb{C}^{M \times n}. \quad (1.2)$$

For a fixed isometry $U \in \mathbb{C}^{n \times d}$ ($U^*U = I_d$) the Gram error is

$$E_{\mathcal{T}} = U^*S^*SU - I_d = \frac{n}{M} \sum_{j \in \mathcal{T}} w_j w_j^* - I_d, \quad w_j^* = e_j^* Q U, \quad \sum_{j=1}^n w_j w_j^* = I_d.$$

The event $\|E_{\mathcal{T}}\| \leq \varepsilon$ is the $(1 \pm \varepsilon)$ subspace-embedding event for range U . The classical (one-round) SRHT of Ailon–Chazelle and Tropp [1, 23] is the case $B = I$ (then $D_y D_x$ is a single sign diagonal). Throughout, $1 \leq d \leq n$ and $q \geq 1$ is an integer; the even trace moment $\mathbb{E} \operatorname{tr} E_{\mathcal{T}}^{2q}$ is the basic object.

Put

$$\begin{aligned} a_q &= \alpha q(2q+1), & b_q &= \beta(d+8q^2) + \alpha q(2q-1), \\ D_q &= a_q + b_q = \beta d + (8\beta + 4\alpha)q^2, & \Lambda_q &= (\sqrt{a_q} + \sqrt{2b_q})^2. \end{aligned} \quad (1.3)$$

For flat transforms $D_q = d + 12q^2$. Always

$$D_q \leq \Lambda_q \leq 3D_q, \quad (1.4)$$

since $\Lambda_q = a_q + 2b_q + 2\sqrt{2a_q b_q} \geq a_q + b_q$ and, by Cauchy–Schwarz, $(\sqrt{a_q} \cdot 1 + \sqrt{b_q} \cdot \sqrt{2})^2 \leq 3(a_q + b_q)$. For sampling we write

$$\rho = \frac{M}{n}, \quad p_{n,M} = \mathbb{P}\{\operatorname{Bin}(n-1, \rho) = M\}, \quad \lambda_{n,M} = \frac{n-M}{M(n-1)} \quad (n \geq 2). \quad (1.5)$$

1.2 Main results

The first result is the canonical sampling prescription. It rests on the retained row-factor estimate and the graded multiplication-operator moment identity, which we import (Section 2).

Theorem A (canonical prescription). *Let $0 < \varepsilon < 1$ and $q \geq 1$. For every integer M with $\min\{n, \lceil 20\Lambda_q/\varepsilon^2 \rceil\} \leq M \leq n$ and every fixed isometry U ,*

$$\mathbb{P}\{\|E_{\mathcal{T}}\| > \varepsilon\} \leq \frac{6}{5} d 4^{-q}.$$

Consequently, for $0 < \delta < 1$,

$$\boxed{q_* = \left\lceil \log_4 \frac{6d}{5\delta} \right\rceil, \quad M = \min\{n, \lceil 20\Lambda_{q_*}/\varepsilon^2 \rceil\}} \quad (1.6)$$

gives $\mathbb{P}\{\|E_{\mathcal{T}}\| > \varepsilon\} \leq \delta$. Each sign family may be only $\min(n, 4q)$ -wise independent, provided the two families and $\mathcal{T}$ are mutually independent. Since $\Lambda_q \leq 3D_q$, the coherence-only prescription $M = \min\{n, \lceil 60D_{q_*}/\varepsilon^2 \rceil\}$ has the same guarantee.

The constant 20 calibrates the *exact* moment root (Theorem 3.3); $72\Lambda_q$ and $216D_q$ calibrate a crude relaxation of it. Section 4.1 proves the ordering $216D_q \geq 72\Lambda_q > 60D_q \geq 20\Lambda_q$ and explains where each factor comes from (Figure 2).

Theorem B (direct selector). *For each prescribed pair (q, M) with $q \geq 1$ and $1 \leq M < n$, the explicit quantity*

$$\operatorname{Cert}(q, M) = d [\mathcal{R}_q(M)/\varepsilon]^{2q}$$

from Theorem 3.3 bounds the embedding failure probability. Algorithm 1 returns only a pair whose certificate is at most δ , or the exact sketch $M = n$. Its guarantee uses $\min(n, 4q)$ -wise independence at the returned order.

The ideal minimum over the searched orders is no larger than the canonical prescription if $q_{\max} \geq q_*$. A bisection implementation is a search heuristic, rather than a proof of that minimum.

In the four examples $d = 10, 10^2, 10^3, 10^4$ at $n = 2^{30}$, $\varepsilon = 1/2$, $\delta = 0.01$, the checked returned widths are respectively

$$58\,406, \quad 125\,760, \quad 346\,325, \quad 1\,423\,405.$$

They are smaller than the canonical widths 102 608, 157 664, 427 412, and 2 192 958.

Theorem D (RP-selected frames). Fix an isometry $V \in \mathbb{C}^{N \times k}$ and, for each k -subset $J \subseteq [N]$, a target isometry U_J , before the sketch is drawn. After drawing S , choose any positive definite $C(S)$ and run k steps of randomly pivoted Cholesky on $VC(S)V^*$; release the unordered pivot set J . If $\Gamma_{C(S)} \leq \bar{\Gamma}$ almost surely, where $\Gamma_C = k! \det C / \prod_{j < k} \tau_j(C) \in [1, k!]$, then the selected frame satisfies $\mathbb{P}\{\|U_J^* S^* S U_J - I_d\| > \varepsilon\} \leq \delta$ with the prescription (1.6) at confidence $\delta/\bar{\Gamma}$. For $k = 2$ the allowance $\delta/2$ always suffices and beats the second-moment allowance $8\delta^2/9$ for every $\delta < 9/16$.

Theorem C (whole-sketch Nyström energy). Let A be flat and B unitary, let the sign families be four-wise independent, and let $K \succeq 0$. With $\mathcal{N}_K(S^*) = KS^*(SKS^*)^\dagger SK$,

$$\mathbb{E}\|K - \mathcal{N}_K(S^*)\|_F^2 \leq \lambda_{n,M}[(\operatorname{tr} K)^2 + \|K\|_F^2].$$

If K is real and A, B are real flat orthogonal matrices (e.g. $A = B = H_n$), the Gram-error energy has the exact value $\lambda_{n,M}[(1 - \frac{2}{n})(\operatorname{tr} K)^2 + (1 - \frac{4}{n})\|K\|_F^2 + \frac{4}{n} \sum_i K_{ii}^2]$ and

$$\boxed{\mathbb{E}\|K - \mathcal{N}_K(S^*)\|_F^2 \leq \frac{2}{M} \left(1 - \frac{M}{n}\right) (\operatorname{tr} K)^2.}$$

The one-round real flat law ($B = I$) has exact Gram-error energy $\lambda_{n,M}[(\operatorname{tr} K)^2 + \|K\|_F^2 - 2 \sum_i K_{ii}^2]$ and satisfies the same boxed bound.

The bound is rank-blind: it is an approximate-matrix-multiplication variance [10] for the product $K^{1/2} \cdot K^{1/2}$, not a relative-error low-rank guarantee. Its virtue is that it counts the actual M rows of one sketch with shared signs and reaches zero at $M = n$. It yields an $O(1/\varepsilon)$ -query trace estimator and an $O(1/\varepsilon)$ -query diagonal estimator (Corollaries 7.7 and 7.9).

Theorem E (fresh-row drift). For flat A , a fixed row j of a freshly drawn Q , and $z = \sqrt{n} Q^* e_j$, every PSD $R_0 \neq 0$ satisfies $\mathbb{E}(\operatorname{tr} R_0 - \operatorname{tr} R_1) \geq \operatorname{tr}(R_0^2)/(\kappa_0 \operatorname{tr} R_0)$ for the Nyström update $R_1 = R_0 - R_0 z (z^* R_0 z)^\dagger z^* R_0$, where $\kappa_0 = e^{2-\gamma}/2 = 2.0743\dots$, with γ Euler's constant. Only one Khintchine factor occurs, not one per transform, and κ_0 equals the constant of an i.i.d. Rademacher probe.

Section 9 places the canonical prescription inside the path-local embedding framework for sketched randomly pivoted Cholesky developed in *RPCholesky under approximate and sketched pivot probabilities* [18].

1.3 Contributions and relation to earlier work

The embedding order is an earlier result. The one-round SRHT and its subspace-embedding analysis are established tools [1, 11, 23, 5]. The present two-round Walsh distribution is the rerandomized transform in Amsel et al. [2, Definition 5.3 and Problem 5.6]. Yang [24, Theorem 1] proves the prescribed constant-failure width $M = \min\{n, \lceil Cd/\varepsilon^2 \rceil\}$ for fixed real frames under this exact two-Walsh law. The earlier repository paper [16] gives a different

argument with allowance $d + 27q^3$; the graded framework [15, Theorem 10.1] gives the quadratic allowance D_q for arbitrary coherent unitary pairs. We claim neither the first two-round embedding theorem nor a new optimal asymptotic row order at constant confidence.

What the present calibration adds. The retained row-factor analysis [14, Theorem 4.1] supplies Λ_q and the density-sensitive Bernoulli moment bound used here. The exact barycenter coupling is also already proved in that analysis and in the graded framework. Their stated prescriptions use the relaxations $c^{-1} \leq 2$ and $1 - \rho \leq 1$. We instead retain $c = 1 - p_{n,M}$ and $\rho = M/n$ together. An elementary binomial-atom estimate then gives the sufficient coefficient $20\Lambda_q$, or $60D_q$, in place of $72\Lambda_q$, or $216D_q$. This is an improvement to these particular sufficient conditions; it does not establish that the constants are optimal, or that they improve every earlier analysis of the Walsh law. Searching the explicit bound over (q, M) recovers further scalar slack. The quadratic confidence allowance remains unchanged.

Whole-sketch consequences and inherited ideas. The finite-population fourth-moment identities in Section 7 apply to all rows of one actual sketch, which share both sign diagonals. They yield a rank-blind Nyström estimate and structured-sketch versions of the sketch-and-correct ideas in Hutch++ [21] and Diag++ [4]; the $O(1/\varepsilon)$ query order and the variance-reduction principle are earlier results. Approximate matrix multiplication provides the relevant error scale [10]. In particular, Nyström++ [26] already combines a Nyström approximation with independent residual probes in one pass. Our trace estimator instantiates that idea with the present without-replacement structured sketch and its explicit second-moment bound. SRHT-based Nyström approximation itself is also established work [3], including block constructions for distributed systems.

The adaptive-frame argument specializes the classical adaptive-to-volume sampling comparison of Deshpande and Vempala [8, Proposition 7]. It has the same fixed-range limitation as the gate used in the repository SparseStack companion [25]. Its contribution here is the explicit SRHT budget obtained by combining that comparison with the canonical prescription. The fresh-row drift is a Khintchine-based specialization of the product-probe argument [19], with the same constant as a Rademacher probe.

Other sampling comparisons. Gross and Nesme [12] establish the classical comparison between sampling with and without replacement for a finite matrix collection. Our auxiliary comparison is between Bernoulli sampling and a fixed-size subset, with a centered conditional barycenter; these are different sampling laws. The general algorithmic role of structured dimension reduction, including weaker injectivity guarantees, is developed by Camaño et al. [6]. No injectivity result is substituted for the two-sided subspace embedding required here.

1.4 Roadmap

Figure 1 shows the logical structure. The imported operator estimates are used only through Imported Theorem 2.1. Sections 3–5 turn it into prescriptions; Section 6 makes the frame adaptive; Sections 7–8 are independent of the moment machinery and use only fourth moments and Khintchine’s inequality; Section 9 connects to sketched pivoting; Section 10 contains all numerical tables.

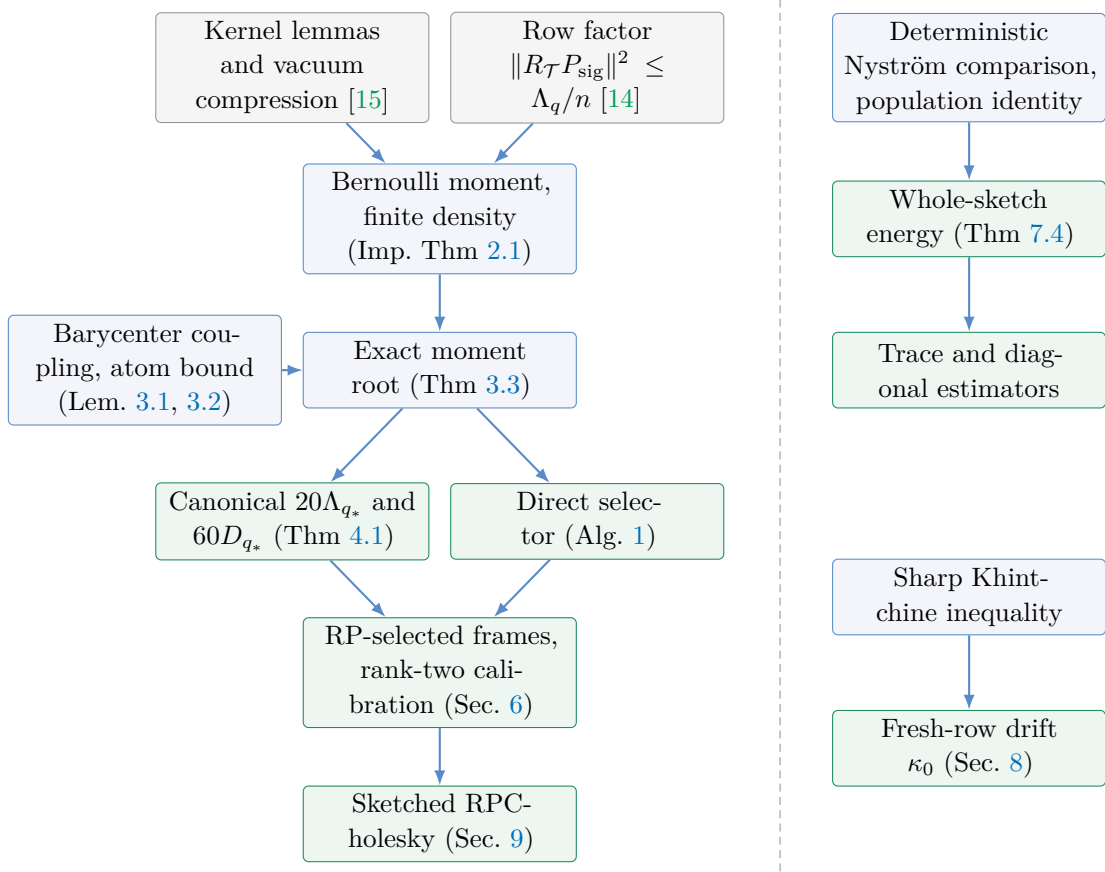


Figure 1: Roadmap. Grey: imported results. Green: results of this paper. The right column uses only fourth moments and Khintchine’s inequality.

2 Imported results and limited independence

The embedding prescriptions are conditional on the following previously proved estimate. We state its complete hypotheses so that readers can use the scalar argument without reconstructing the underlying operator notation. Its proof is in the repository row-factor paper [14], with the coefficient kernels supplied by the graded framework [15]. Neither this section nor the numerical checks below constitute a new verification of all those imported kernels.

Let $(\eta_j)_{j \leq n}$ be independent Bernoulli(ρ) selectors, independent of the signs, $0 < \rho < 1$, $m = \rho n$, and

$$E_\rho = \rho^{-1} W^* \text{Diag}(\eta) W - I_d = \sum_j (\eta_j / \rho - 1) w_j w_j^*, \quad W = QU.$$

Imported theorem 2.1 (Retained row factor and Bernoulli moments; [14], building on [15]). *Under fully independent signs and selectors, for every integer $q \geq 1$,*

$$\mathbb{E} \text{tr} E_\rho^{2q} \leq d \left[2 \sqrt{\frac{(1-\rho)\Lambda_q}{m}} + |1-2\rho| \frac{\Lambda_q}{m} \right]^{2q}. \quad (2.1)$$

Remark 2.2 (Provenance of (2.1)). In [15] the error E_ρ is represented as a multiplication operator on the finite product space of sign monomials and selector monomials, and $\mathbb{E} \text{tr} E_\rho^{2q} = \sum_{a \leq d} \|M_{\text{cut}}^q(e_a \otimes \Omega)\|^2$ exactly, where M_{cut} is the multiplication operator compressed to sign grades

at most $2q$ and selector grade at most q (one multiplication moves each sign grade by at most two and the selector grade by at most one). In the normalized basis of a selector site, multiplication by $\eta/\rho - 1$ is $\begin{pmatrix} 0 & \sigma \\ \sigma & \tau \end{pmatrix}$ with $\sigma = \sqrt{(1-\rho)/\rho}$, $\tau = (1-2\rho)/\rho$, and the positive-selector lemma of [15] gives $\|M_{\text{cut}}\| \leq 2\sigma\sqrt{L_q} + |\tau|L_q$, where L_q is the largest norm of a sum of at most q compressed positive row effects. The retained row-factor theorem of [14] bounds the row map before forming its Gram, $\|R_{\mathcal{T}}P_{\text{sig}}\|^2 \leq \Lambda_q/n$ for $|\mathcal{T}| \leq q$, so $L_q \leq \min\{1, \Lambda_q/n\}$; substituting gives (2.1). That theorem uses the joint-creation, mixed-channel and collective double-annihilation kernel bounds of [15]. The earlier bound $L_q \leq 9D_q/n$ of [15] gives the same statement with Λ_q replaced by $9D_q$.

Lemma 2.3 (Limited independence). *Every entry of $E_{\mathcal{T}}$ and of E_{ρ} is a polynomial of degree at most two in each sign family. Hence $\mathbb{E} \text{tr} E_{\mathcal{T}}^{2q}$ is unchanged if each sign family is only $\min(n, 4q)$ -wise independent and unbiased, provided the two families and the subset (or the selectors) are mutually independent.*

Proof. $w_j w_j^*$ has entries $\sum_{a,b,i,l} A_{ja} \overline{A_{jb}} B_{ai} \overline{B_{bl}} y_a y_b x_i x_l \overline{U_i} \overline{U_l}$, of degree two in x and in y . Expanding $\text{tr} E_{\mathcal{T}}^{2q}$ produces a linear combination of monomials, each involving at most $4q$ distinct coordinates of either family. The expectation of such a monomial depends only on the joint law of at most $\min(n, 4q)$ coordinates of each family, which is the uniform law on signs under $\min(n, 4q)$ -wise independence. Conditioning on the subset, which is independent of the signs, completes the proof. $\square$

3 The exact moment root

3.1 From Bernoulli to fixed-size sampling

Fix Hermitian matrices $H_1, \dots, H_n$ and $1 \leq M < n$, $\rho = M/n$. For a Bernoulli(ρ) subset $\mathcal{B}$ and a uniform M -subset $\mathcal{T}$ put

$$Z_{\mathcal{B}} = \sum_j (\mathbf{1}_{j \in \mathcal{B}} - \rho) H_j, \quad Z_{\mathcal{T}} = \sum_j (\mathbf{1}_{j \in \mathcal{T}} - \rho) H_j.$$

Lemma 3.1 (Exact barycenter coupling). *There is a coupling of $\mathcal{B}$ and $\mathcal{T}$ such that*

$$\mathbb{E}[Z_{\mathcal{B}} \mid \mathcal{T}] = c Z_{\mathcal{T}}, \quad c = 1 - p_{n,M} \geq \frac{1}{2}.$$

Consequently, for every integer $q \geq 1$, $\mathbb{E} \text{tr} Z_{\mathcal{T}}^{2q} \leq c^{-2q} \mathbb{E} \text{tr} Z_{\mathcal{B}}^{2q}$.

Proof. Draw $\mathcal{B}$, with size $K \sim \text{Bin}(n, \rho)$. If $K > M$ delete $K - M$ uniformly chosen elements; if $K < M$ add $M - K$ uniformly chosen missing elements. By symmetry $\mathcal{T}$ is a uniform M -subset (independent of K), and conditionally on $\mathcal{T}$ the law of $\mathcal{B}$ is invariant under permutations fixing $\mathcal{T}$. Hence $\mathbb{P}(j \in \mathcal{B} \mid \mathcal{T})$ takes one value θ_1 for $j \in \mathcal{T}$ and one value θ_0 for $j \notin \mathcal{T}$. An element of $\mathcal{T}$ is missing from $\mathcal{B}$ only if it was added, and an element outside $\mathcal{T}$ lies in $\mathcal{B}$ only if it was deleted, so with $a = \mathbb{E}(K - M)_+ = \mathbb{E}(M - K)_+$ (equal because $\mathbb{E}K = M$),

$$\theta_1 = 1 - \frac{a}{M}, \quad \theta_0 = \frac{a}{n - M}.$$

With $v = n\rho(1-\rho) = M(1-\rho) = \rho(n-M)$ we get $\theta_1 - \rho = (1-\rho)(1-a/v)$ and $\theta_0 - \rho = -\rho(1-a/v)$, so $\mathbb{E}[\mathbf{1}_{j \in \mathcal{B}} - \rho \mid \mathcal{T}] = (1-a/v)(\mathbf{1}_{j \in \mathcal{T}} - \rho)$ and $\mathbb{E}[Z_{\mathcal{B}} \mid \mathcal{T}] = (1-a/v)Z_{\mathcal{T}}$.

Let $Y \sim \text{Bin}(n-1, \rho)$ and write $K = Y + \xi$ with an independent Bernoulli(ρ) variable ξ . The identity $k \binom{n}{k} = n \binom{n-1}{k-1}$ gives $\mathbb{E}[K \mathbf{1}_{K \geq M+1}] = n\rho \mathbb{P}(Y \geq M) = M \mathbb{P}(Y \geq M)$, and $\mathbb{P}(K \geq M+1) = \mathbb{P}(Y \geq M+1) + \rho \mathbb{P}(Y = M)$. Therefore

$$a = \mathbb{E}[K \mathbf{1}_{K \geq M+1}] - M \mathbb{P}(K \geq M+1) = M(1-\rho) \mathbb{P}(Y = M) = v p_{n,M},$$

so $1 - a/v = 1 - p_{n,M} = c$. The atoms $\mathbb{P}(Y = M)$ and $\mathbb{P}(Y = M-1)$ have ratio $\frac{n-M}{M} \cdot \frac{\rho}{1-\rho} = 1$, so each is at most $1/2$ and $c \geq 1/2$.

Finally $Z \mapsto \text{tr } Z^{2q} = \|Z\|_{S_{2q}}^{2q}$ is convex on Hermitian matrices, and conditional Jensen gives $\mathbb{E} \text{tr } Z_{\mathcal{B}}^{2q} \geq \mathbb{E} \text{tr}(\mathbb{E}[Z_{\mathcal{B}} | \mathcal{T}])^{2q} = c^{2q} \mathbb{E} \text{tr } Z_{\mathcal{T}}^{2q}$. $\square$

Lemma 3.2 (Binomial atom). *For $1 \leq M < n$, $\rho = M/n$ and $v = M(1-\rho)$,*

$$p_{n,M} = \mathbb{P}\{\text{Bin}(n-1, \rho) = M\} = \mathbb{P}\{\text{Bin}(n, \rho) = M\} \leq \min\left\{\frac{1}{2}, \sqrt{\frac{\pi}{8v}}\right\}.$$

Proof. The equality follows from $\binom{n-1}{M} = (1-\rho)\binom{n}{M}$, and the bound $1/2$ was shown in Lemma 3.1. By Fourier inversion,

$$\mathbb{P}\{\text{Bin}(n, \rho) = M\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} (1-\rho + \rho e^{it})^n e^{-iMt} dt \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |1-\rho + \rho e^{it}|^n dt.$$

Now $|1-\rho + \rho e^{it}|^2 = 1 - 4\rho(1-\rho) \sin^2(t/2) \leq \exp\{-4\rho(1-\rho) \sin^2(t/2)\}$, so the integrand is at most $\exp\{-2v \sin^2(t/2)\}$. On $[-\pi, \pi]$ concavity of the sine gives $|\sin(t/2)| \geq |t|/\pi$. Extending the Gaussian integral to $\mathbb{R}$,

$$p_{n,M} \leq \frac{1}{2\pi} \int_{\mathbb{R}} e^{-2vt^2/\pi^2} dt = \frac{1}{2\pi} \cdot \pi \sqrt{\frac{\pi}{2v}} = \sqrt{\frac{\pi}{8v}}. \quad \square$$

3.2 The root

Theorem 3.3 (Exact moment root). *Let $1 \leq M < n$ and $\rho = M/n$. For every integer $q \geq 1$ and every fixed isometry U ,*

$$\mathbb{E} \text{tr } E_{\mathcal{T}}^{2q} \leq d \mathcal{R}_q(M)^{2q}, \quad \mathcal{R}_q(M) = \frac{2\sqrt{(1-\rho)\Lambda_q/M} + |1-2\rho| \Lambda_q/M}{1 - p_{n,M}}. \quad (3.1)$$

In particular, for every $\varepsilon > 0$,

$$\mathbb{P}\{\|E_{\mathcal{T}}\| > \varepsilon\} \leq \text{Cert}(q, M) := d \left(\frac{\mathcal{R}_q(M)}{\varepsilon} \right)^{2q}. \quad (3.2)$$

Both statements hold under $\min(n, 4q)$ -wise independent signs. For $M = n$ the error vanishes identically.

Proof. Condition on the signs and apply Lemma 3.1 to $H_j = w_j w_j^*$. Since $\sum_j w_j w_j^* = I_d$, we have $E_{\mathcal{T}} = \rho^{-1} Z_{\mathcal{T}}$ and $E_{\rho} = \rho^{-1} Z_{\mathcal{B}}$ with $m = \rho n = M$. Averaging over the signs, $\mathbb{E} \text{tr } E_{\mathcal{T}}^{2q} \leq c^{-2q} \mathbb{E} \text{tr } E_{\rho}^{2q}$, and Imported Theorem 2.1 with $m = M$ gives (3.1). Since $\|E\|^{2q} \leq \text{tr } E^{2q}$ for Hermitian E , Markov's inequality gives (3.2). Limited independence follows from Lemma 2.3. If $M = n$ then $S^* S = Q^* Q = I_n$. $\square$

Remark 3.4 (The crude root). Using $c^{-1} \leq 2$, $1-\rho \leq 1$ and $|1-2\rho| \leq 1$ separately gives

$$\mathcal{R}_q(M) \leq \mathcal{R}_q^{\text{cr}}(M) := 4\sqrt{\Lambda_q/M} + 2\Lambda_q/M \leq 4\sqrt{3D_q/M} + 6D_q/M. \quad (3.3)$$

These are the roots calibrated in [14] by $72\Lambda_q$ and $216D_q$. The root $\mathcal{R}_q$ couples the three quantities: when ρ is close to one, c can be close to $1/2$ but then $1 - \rho$ is small; when $1 - \rho$ is not small, the atom $p_{n,M}$ is of order $M^{-1/2}$ and c is close to one.

Remark 3.5 (No lower bound on q). Imported Theorem 2.1, Lemma 3.1, Lemma 2.3 and therefore Theorem 3.3 hold for *every* integer $q \geq 1$. The order q is tied to d and δ only through Markov's inequality. In particular the convention $q = \lceil \log_4(4d/\delta) \rceil$ of [15, 14], which yields failure $\delta/4$, has no structural role.

4 The canonical prescription

Theorem 4.1 (Coefficient 20 on Λ_q). *Let $0 < \varepsilon < 1$ and $q \geq 1$. If $\min\{n, \lceil 20\Lambda_q/\varepsilon^2 \rceil\} \leq M \leq n$, then for every fixed isometry U*

$$\mathbb{E} \operatorname{tr} E_{\mathcal{T}}^{2q} \leq \frac{6}{5} d (\varepsilon/2)^{2q}, \quad \mathbb{P}\{\|E_{\mathcal{T}}\| > \varepsilon\} \leq \frac{6}{5} d 4^{-q}. \quad (4.1)$$

With $q = q_ = \lceil \log_4(6d/(5\delta)) \rceil$, $0 < \delta < 1$, the failure probability is at most δ . The signs may be $\min(n, 4q)$ -wise independent.*

Proof. If $M = n$ there is nothing to prove, so let $M < n$; then $M \geq 20\Lambda_q/\varepsilon^2$. Put $a = \sqrt{\Lambda_q/M} \leq \varepsilon/\sqrt{20}$ and $r = 1 - \rho \in (0, 1)$. Since $|1 - 2\rho| \leq 1$, the numerator of $\mathcal{R}_q(M)$ is at most $2\sqrt{r}a + a^2$.

Small missing fraction: $r \leq 1/5$. The simple lower bound $c \geq 1/2$ is sufficient because the square-root term already contains the small missing fraction. It gives

$$\mathcal{R}_q(M) \leq 2(2\sqrt{1/5} \varepsilon/\sqrt{20} + \varepsilon^2/20) = \frac{2}{5}\varepsilon + \frac{1}{10}\varepsilon^2 \leq \frac{1}{2}\varepsilon.$$

Larger missing fraction: $r \geq 1/5$. Here the large prescribed width makes the binomial atom small. Since $\alpha, \beta, d \geq 1$, $b_q \geq 10q^2 - q + 1 \geq 9q^2$, hence $\Lambda_q \geq 2b_q \geq 18q^2$ and $M \geq 20\Lambda_q > 360q^2$. Lemma 3.2 with $v = Mr \geq M/5$ gives

$$p_{n,M} \leq \sqrt{\frac{5\pi}{8M}} < \sqrt{\frac{5\pi}{8 \cdot 360q^2}} = \frac{\sqrt{\pi}}{24q} < \frac{1}{12q}.$$

Therefore, using $\varepsilon^2 \leq \varepsilon$,

$$\mathcal{R}_q(M) \leq \frac{2a + a^2}{1 - 1/(12q)} \leq \frac{\varepsilon(2/\sqrt{20} + 1/20)}{1 - 1/(12q)} < \frac{\varepsilon/2}{1 - 1/(12q)},$$

where $2/\sqrt{20} + 1/20 < 1/2$ is equivalent to $4/20 < (9/20)^2$, i.e. $80 < 81$. Bernoulli's inequality gives $(1 - 1/(12q))^{2q} \geq 1 - 2q/(12q) = 5/6$.

In both cases $\mathcal{R}_q(M)^{2q} \leq \frac{6}{5}(\varepsilon/2)^{2q}$, and Theorem 3.3 gives (4.1). The choice of q_* gives $\frac{6}{5}d4^{-q_*} \leq \delta$. $\square$

Corollary 4.2 (Coherence-only form). *The conclusions of Theorem 4.1 hold for every $M \geq \min\{n, \lceil 60D_q/\varepsilon^2 \rceil\}$, in particular for $q_* = \lceil \log_4(6d/(5\delta)) \rceil$ and $M = \min\{n, \lceil 60D_{q_*}/\varepsilon^2 \rceil\}$. For flat transforms $D_q = d + 12q^2$.*

Proof. $60D_q \geq 20\Lambda_q$ by (1.4). $\square$

Corollary 4.3 (Rank-only coefficient at confidence 0.99). *For flat transforms, $0 < \varepsilon < 1$ and every fixed d -dimensional subspace, $M = \min\{n, \lceil 11580 d/\varepsilon^2 \rceil\}$ rows give failure probability at most 0.01.*

Proof. At $\delta = 0.01$, $q_* = \lceil \log_4(120d) \rceil \geq 4$. We claim $q_*^2 \leq 16d$. For $q_* = 4$ this is clear. If $q_* = k \geq 5$, then $120d > 4^{k-1}$, so $k^2/d < 120k^2/4^{k-1}$; the right side decreases for $k \geq 5$ (consecutive ratio $(k+1)^2/(4k^2) < 1$) and equals $3000/256 < 16$ at $k = 5$. Hence $D_{q_*} = d + 12q_*^2 \leq 193d$, and $60 \cdot 193 = 11580$; apply Corollary 4.2, which allows every larger M . $\square$

The printed rank-only coefficients for the same model were 308 224 in [15] and 65 016 in [14]. Numerically, $\sup_{d \leq 3 \cdot 10^5} 20\Lambda_{q_*}/d = 11\,252.8$, attained at $d = 1$.

Corollary 4.4 (Crude-root prescriptions at the reduced order). *Let $0 < \varepsilon < 1$, $0 < \delta < 1$ and $q_f = \lceil \log_4(d/\delta) \rceil$. Then $M = \min\{n, \lceil 72\Lambda_{q_f}/\varepsilon^2 \rceil\}$, and a fortiori $M = \min\{n, \lceil 216D_{q_f}/\varepsilon^2 \rceil\}$, give failure probability at most δ under $\min(n, 4q_f)$ -wise independent signs.*

Proof. Since $d \geq 1 > \delta$, $q_f \geq 1$. For $M < n$, the crude root (3.3) satisfies $\mathcal{R}_{q_f}^{\text{cr}}(M) \leq \varepsilon(4/\sqrt{72} + 2\varepsilon/72) = \varepsilon(\sqrt{2}/3 + \varepsilon/36) < \varepsilon/2$, because $\sqrt{2}/3 + 1/36 < 1/2$ is $12\sqrt{2} < 17$, i.e. $288 < 289$. Theorem 3.3 and Remark 3.5 give failure at most $d4^{-q_f} \leq \delta$. $\square$

Corollary 4.4 settles whether the order $q = \lceil \log_4(4d/\delta) \rceil = q_f + 1$ used in [15, 14] is needed: it is not, and dropping one order saves up to 36% of the $72\Lambda_q$ rows at small d (Table 2). The canonical order q_* already lies in $\{q_f, q_f + 1\}$, so this saving is essentially built into Theorem 4.1.

4.1 The chain of constants

Proposition 4.5 (Two roots, two allowances). *For every $q \geq 1$ and all parameters,*

$$216D_q \geq 72\Lambda_q \geq 72D_q > 60D_q \geq 20\Lambda_q, \quad (4.2)$$

where the first and last inequalities are equalities only if $b_q = 2a_q$; for flat transforms all inequalities are strict. The first two prescriptions calibrate the crude root $\mathcal{R}_q^{\text{cr}}$ (failure $\leq d4^{-q}$); the last two calibrate the exact root $\mathcal{R}_q$ (failure $\leq \frac{6}{5}d4^{-q}$). For flat transforms and $q \geq 3$, moreover,

$$60D_{q+1} < 72\Lambda_q, \quad (4.3)$$

so whenever $d/\delta > 16$ the coherence-only exact-root prescription at $q_ \leq q_f + 1$ is smaller than the crude-root prescription $72\Lambda_{q_f}$ at its own best order.*

Proof. The chain is (1.4) multiplied by 72 and 20. Equality in the Cauchy–Schwarz step $\Lambda_q \leq 3D_q$ requires $\sqrt{a_q} : \sqrt{b_q} = 1 : \sqrt{2}$, i.e. $b_q = 2a_q$, i.e. $\beta(d + 8q^2) = \alpha q(2q + 3)$; for $\alpha = \beta = 1$ this would mean $d = -6q^2 + 3q < 0$. The calibrations are Theorem 4.1, Corollary 4.2 and the proof of Corollary 4.4. For flat transforms $\Lambda_q \geq a_q + 2b_q = 2d + 22q^2 - q$, so $72\Lambda_q - 60D_{q+1} \geq 84d + 864q^2 - 1512q - 720$, which increases in $q \geq 1$ and is positive at $q = 3$. Finally $q_f \geq 3$ iff $d/\delta > 16$, and $q_* \leq q_f + 1$. $\square$

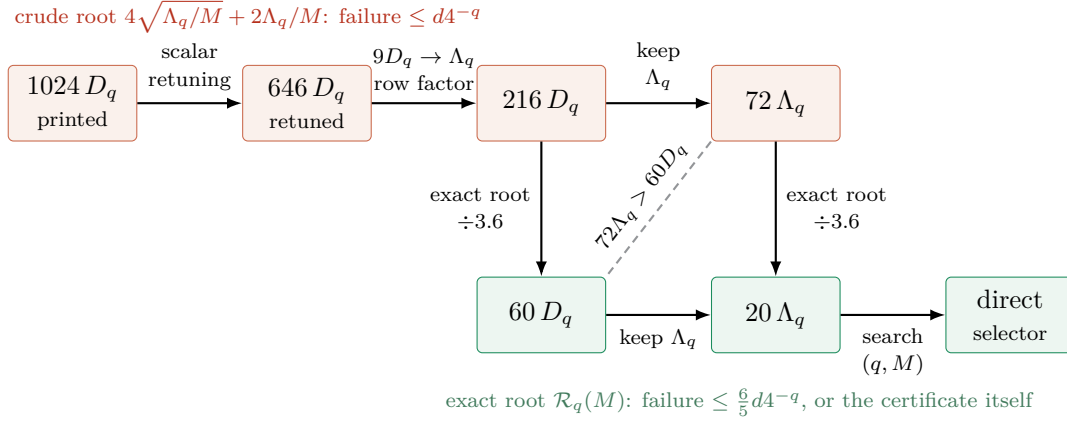


Figure 2: The chain of sufficient row coefficients for the same two-round fixed-size law. Top row: calibrations of the crude root. Bottom row: calibrations of the exact root (3.1). Each vertical arrow divides the coefficient by 3.6; the dashed comparison is (4.2).

Reading the chain. The two mechanisms are independent and multiply.

- *Allowance.* Bounding the row map before forming its Gram gives $L_q \leq \Lambda_q/n$; the relaxation $\Lambda_q \leq 3D_q$ trades an explicit formula for a factor at most three. This is $216 = 3 \cdot 72$ and $60 = 3 \cdot 20$.
- *Root.* The crude root doubles the linear term ($c^{-1} \leq 2$ and $1 - \rho \leq 1$) and needs $4/\sqrt{C} + 2/C \leq 1/2$, whose threshold is $C \approx 71.8$. The exact root needs only $2/\sqrt{C} + 1/C \leq 1/2$, threshold $C \approx 19.8$; the atom is paid by the factor $6/5$ in the failure bound. This is $72 \rightarrow 20$ and $216 \rightarrow 60$.

Consequently $72\Lambda_q$, which keeps the *smaller* allowance but the *worse* root, loses to $60D_q$, which keeps the better root: the root gains 3.6 while the allowance loses at most 3. Figure 2 summarizes the history, including the printed coefficient 1024 of [15] and its pure scalar retuning 646 (allowance $9D_q$, crude root). The same exact-root argument applied to the allowance $9D_q$ gives the coefficient 180.

5 The direct deterministic selector

The certificate (3.2) is an explicit function of (q, M) and of the known parameters. Nothing forces the root to be below $\varepsilon/2$ or q to equal q_* ; any pair with $\text{Cert}(q, M) \leq \delta$ is a valid prescription. Algorithm 1 chooses the pair before any randomness is drawn.

The canonical starting point prevents the heuristic implementation from returning a larger width than the canonical prescription when $q_{\max} \geq q_*$. It does not certify that the returned width is globally minimal. For a guaranteed minimum one must enumerate all candidate widths or prove an appropriate monotonicity statement.

The atom $p_{n,M}$ should be evaluated through $\log \Gamma$ (or Stirling's series) in high precision; the certificate is best handled as $\log \text{Cert} = \log d + 2q \log(\mathcal{R}_q(M)/\varepsilon)$.

Proposition 5.1 (Validity and domination). *(i) If Algorithm 1 returns (q, M) with $M < n$, then $\mathbb{P}\{\|E_{\mathcal{T}}\| > \varepsilon\} \leq \text{Cert}(q, M) \leq \delta$ for every fixed isometry U , under $\min(n, 4q)$ -wise*

Algorithm 1 Direct selector for the two-round sketch

Require: $n, d, \varepsilon, \delta, \alpha, \beta$; maximal order $q_{\max}$; safety margin $s \geq 0$

```
1: best  $\leftarrow (\perp, n)$   $\triangleright M = n$  is always exact
2: if  $q_{\max} \geq q_*$  then
3:    $M_0 \leftarrow \min\{n, \lceil 20\Lambda_{q_*}/\varepsilon^2 \rceil\}$ 
4:   best  $\leftarrow (q_*, M_0)$   $\triangleright$  proved canonical starting point
5: end if
6: for  $q = 1, \dots, q_{\max}$  do
7:   compute  $\Lambda_q$  from (1.3)
8:   if  $\text{Cert}(q, n-1) \leq \delta$  then
9:      $M_q \leftarrow$  candidate in  $[1, n-1]$  found by bisection  $\triangleright$  heuristic search
10:     $M' \leftarrow \min\{M_q + s, n-1\}$ 
11:    if  $\text{Cert}(q, M') \leq \delta$  and  $M' < \text{best}.M$  then  $\triangleright$  re-check at the final  $M$ 
12:      best  $\leftarrow (q, M')$ 
13:    end if
14:  end if
15: end for
16: if best. $M = n$  then
17:   return  $M = n$ , certificate 0, no sign-independence requirement
18: end if
19: return best =  $(q, M)$ ,  $\text{Cert}(q, M)$ , and independence level  $\min(n, 4q)$ 
```

independent signs. If it returns $M = n$ the embedding is exact. Validity does not depend on the search being exact.

(ii) Let $M^* = \min_{q \leq q_{\max}} \min\{M < n : \text{Cert}(q, M) \leq \delta\}$ (or n if this set is empty). If $q_{\max} \geq q_*$, then M^* is at most each of $\lceil 20\Lambda_{q_*}/\varepsilon^2 \rceil$ and $\lceil 60D_{q_*}/\varepsilon^2 \rceil$, and at most $\lceil 72\Lambda_q/\varepsilon^2 \rceil$ and $\lceil 216D_q/\varepsilon^2 \rceil$ for every $q \leq q_{\max}$ with $d4^{-q} \leq \delta$, whenever these are below n .

Proof. (i) is Theorem 3.3, since the certificate is evaluated at the returned M . (ii) For $M = \lceil 20\Lambda_{q_*}/\varepsilon^2 \rceil < n$ the proof of Theorem 4.1 shows $\mathcal{R}_{q_*}(M)^{2q_*} \leq \frac{6}{5}(\varepsilon/2)^{2q_*}$, so $\text{Cert}(q_*, M) \leq \frac{6}{5}d4^{-q_*} \leq \delta$; the same holds for $60D_{q_*}$. For $72\Lambda_q$ and $216D_q$, $\mathcal{R}_q \leq \mathcal{R}_q^{\text{cr}} \leq \varepsilon/2$ by the proof of Corollary 4.4, so $\text{Cert}(q, M) \leq d4^{-q} \leq \delta$. Each is therefore a feasible point of the minimization defining M^* . $\square$

Remark 5.2 (Implementation). The stated root $\mathcal{R}_q(M)$ is not monotone in general. For example, take $n = 4$, $d = 1$, $q = 1$ and flat factors, so $\Lambda_1 = 25 + 2\sqrt{66} > 36$. Direct substitution gives

$$\mathcal{R}_1(2) = \frac{8}{5}\sqrt{\Lambda_1}, \quad \mathcal{R}_1(3) = \frac{64}{37} \left(\sqrt{\Lambda_1/3} + \Lambda_1/6 \right) > \mathcal{R}_1(2).$$

Indeed $\Lambda_1/6 > \sqrt{\Lambda_1}$ and $64/37 > 8/5$. Both values in this example are outside the useful certification range; it disproves global monotonicity without locating a nonmonotone feasible interval. A returned M must therefore not be rounded up (for instance to a block size) without re-evaluating Cert at the new value, and bisection is only a search heuristic; the final certificate check makes every newly accepted candidate valid. The selected order may exceed q_* ; under limited independence the sign families must then be $\min(n, 4q)$ -wise independent (72-wise for $q = 18$). Fully independent signs are unaffected.

At $n = 2^{30}$, $\varepsilon = 1/2$, $\delta = 0.01$ and flat transforms, with $s = 10$, the selector returns

$$(q, M) = (4, 58\,406), (6, 125\,760), (10, 346\,325), (18, 1\,423\,405) \quad \text{for } d = 10, 10^2, 10^3, 10^4,$$

with certificates between 0.0099926 and 0.0099986. The least feasible values are 58 396, 125 750, 346 315, 1 423 395 (numerical; see Section 10). At $d = 10^4$ this is 35% below $20\Lambda_{q^*}$ and 48% below $60D_{q^*}$.

6 Adaptive frames selected by randomly pivoted Cholesky

A fixed-subspace guarantee does not apply to a subspace chosen after looking at the sketch. We allow one specific, useful form of dependence: a frame is selected from a dictionary fixed in advance by running randomly pivoted Cholesky (RP) [7] at full rank on a matrix that may depend on the sketch arbitrarily, provided its range is fixed.

Setting. Fix an isometry $V \in \mathbb{C}^{N \times k}$ and, for every k -subset $J \subseteq [N]$ with $\det V_J \neq 0$ (V_J the rows of V in J), a target isometry $U_J \in \mathbb{C}^{n \times d}$, all before the sketch S is drawn. Let $C(S) \succ 0$ be any measurable $k \times k$ matrix. Given S , run k steps of RP on $A_{\text{aux}} = VC(S)V^*$ with fresh randomness: at a residual R , pick the pivot i with probability $R_{ii}/\text{tr } R$ and replace R by its Schur complement. Release only the unordered pivot set J , and use the frame U_J . The *volume law* $\nu_V(J) = |\det V_J|^2$ is a probability distribution by Cauchy–Binet and does not depend on S . For eigenvalues $\lambda_1 \geq \dots \geq \lambda_k > 0$ of C put

$$\tau_j(C) = \sum_{i>j} \lambda_i, \quad \Gamma_C = \frac{k! \det C}{\prod_{j=0}^{k-1} \tau_j(C)}. \quad (6.1)$$

Proposition 6.1 (Likelihood price of the gate). *Conditionally on S , the released law $\mu_{C(S)}$ satisfies $\mu_{C(S)}(J) \leq \Gamma_{C(S)} \nu_V(J)$ for every J , and $1 \leq \Gamma_C \leq k!$. If every fixed U_J has sketch failure probability at most δ_0 and $\Gamma_{C(S)} \leq \bar{\Gamma}$ almost surely, then the selected frame fails with probability at most $\bar{\Gamma}\delta_0$.*

Proof. **The pivot numerator is a determinant.** An ordered pivot sequence $(j_1, \dots, j_k)$ with set J has probability

$$\prod_{t=1}^k \frac{(R_{t-1})_{j_t j_t}}{\text{tr } R_{t-1}}.$$

Successive Schur complements give

$$\prod_{t=1}^k (R_{t-1})_{j_t j_t} = \det(A_{\text{aux}})_{JJ} = \det C |\det V_J|^2.$$

Each denominator dominates a spectral tail. For the traces, $A_{\text{aux}} - R_t$ is the Nyström approximation on t columns, a PSD matrix of rank at most t dominated by A_{aux} ; with P the projector onto its range, $\text{tr}(A_{\text{aux}} - R_t) = \text{tr } P(A_{\text{aux}} - R_t)P \leq \text{tr } PA_{\text{aux}}P \leq \lambda_1 + \dots + \lambda_t$ by Ky Fan, since A_{aux} has eigenvalues $\lambda_1, \dots, \lambda_k$ and zeros. Hence $\text{tr } R_t \geq \tau_t(C)$, and every ordered sequence has probability at most $\det C \nu_V(J) / \prod_t \tau_t(C)$. Summing over the $k!$ orders gives $\mu_C(J) \leq \Gamma_C \nu_V(J)$. The range of the price follows because $\lambda_{t+1} \leq \tau_t(C) \leq (k-t)\lambda_{t+1}$, multiplying over t gives $\det C \leq \prod_t \tau_t \leq k! \det C$, i.e. $1 \leq \Gamma_C \leq k!$.

Transfer the fixed-frame guarantee. Let B_J be the failure event of the fixed frame U_J , an event in the sketch randomness. Then

$$\mathbb{P}\{B_J \text{ for the selected } J\} = \mathbb{E}_S \sum_J \mu_{C(S)}(J) \mathbf{1}_{B_J}(S) \leq \bar{\Gamma} \sum_J \nu_V(J) \mathbb{P}_S(B_J) \leq \bar{\Gamma} \delta_0,$$

using that ν_V and the frames were fixed before S . $\square$

The factorial comparison is Proposition 7 of Deshpande and Vempala [8]; Γ_C retains its denominators. The same gate, with the same price (written Λ_C there), is used for sparse sketches in [25]; we write Γ_C to avoid a clash with the allowance Λ_q . Nothing in Proposition 6.1 depends on the sketch; only the fixed-frame guarantee it is combined with does. Flat C gives $\Gamma_C = 1$ (full-rank RP on VV^* is volume sampling [9]). For a spike $C = I + \eta uu^*$ ($\|u\| = 1$), $\Gamma_C = k(1 + \eta)/(k + \eta)$. If the eigenvalues are split into contiguous bands of sizes n_ℓ with largest-to-smallest ratio at most ϱ_ℓ inside each band, bounding each tail from below by the remaining eigenvalues of its own band gives

$$\Gamma_C \leq \min\left\{k!, \frac{k!}{\prod_\ell n_\ell!} \prod_\ell \varrho_\ell^{n_\ell}\right\}. \quad (6.2)$$

Corollary 6.2 (SRHT on an RP-selected frame). *In the setting above, let $0 < \varepsilon < 1$, $0 < \delta < 1$, and let $\bar{\Gamma}$ be a deterministic bound for $\Gamma_{C(S)}$ ($\bar{\Gamma} = k!$ is always allowed). Then*

$$q = \left\lceil \log_4 \frac{6d\bar{\Gamma}}{5\delta} \right\rceil, \quad M = \min\{n, \lceil 20\Lambda_q/\varepsilon^2 \rceil\}$$

give $\mathbb{P}\{\|U_J^* S^* S U_J - I_d\| > \varepsilon\} \leq \delta$ for the selected frame, with $\min(n, 4q)$ -wise independent signs. Algorithm 1 run at confidence $\delta/\bar{\Gamma}$ may be used instead.

Proof. Theorem 4.1 with $\delta_0 = \delta/\bar{\Gamma}$ and Proposition 6.1. When $M = n$ every frame is exact. $\square$

Compared with the union bound over the dictionary, whose price can be $\binom{N}{k}$, the gate pays $\log \bar{\Gamma} \leq \log k!$ inside q , independently of N . For flat transforms this gives $M = O(\varepsilon^{-2}[d + (\log(d/\delta) + \log \bar{\Gamma})^2])$. Table 3 evaluates the example $d = 100$, $k = 10$, $N = 1000$.

Proposition 6.3 (Rank-two calibration). *For $k = 2$, $\Gamma_C = 2\lambda_1/(\lambda_1 + \lambda_2) < 2$ for every C , so $\delta_0 = \delta/2$ is always sufficient; with the canonical prescription this is $q = \lceil \log_4(12d/(5\delta)) \rceil$. A second-moment certificate $\mathbb{E}_{\nu_V}(\mu_C/\nu_V)^2 \leq 9/8$ would give, by Cauchy–Schwarz, failure at most $\sqrt{9/8}\sqrt{\delta_0}$, i.e. the allowance $\delta_0 = 8\delta^2/9$. The pointwise allowance is larger whenever $\delta < 9/16$, so the second-moment route is never useful in this range. For $k = 1$ the gate equals the volume law and costs nothing.*

Proof. For $k = 2$, $\Gamma_C = 2\lambda_1\lambda_2/((\lambda_1 + \lambda_2)\lambda_2)$. The Cauchy–Schwarz bound is $\mathbb{E}_\nu[(\mu/\nu)\mathbf{1}_B] \leq (\mathbb{E}_\nu(\mu/\nu)^2)^{1/2} \mathbb{P}_\nu(B)^{1/2}$. Finally $\delta/2 > 8\delta^2/9$ iff $\delta < 9/16$. $\square$

Remark 6.4 (Scope of the gate). The range of A_{aux} must be fixed: if it could move with S , rank-one inputs $e_{j(S)} e_{j(S)}^*$ would encode any label. The pivot order must not be released, since it can carry extra information. The rank must be full, because at partial rank the volume reference depends on C . The frame dictionary must be fixed in advance. The bound $\bar{\Gamma}$ must hold uniformly; inserting the observed $\Gamma_{C(S)}$ into a budget chosen after seeing the sketch is not covered.

7 Whole-sketch Nyström energy

Throughout this section $K \in \mathbb{C}^{n \times n}$ is Hermitian positive semidefinite, $G = S^*S$ and

$$\mathcal{N}_K(S^*) = KS^*(SKS^*)^\dagger SK, \quad R = K - \mathcal{N}_K(S^*).$$

Rescaling S by a scalar does not change $\mathcal{N}_K(S^*)$, but it changes G ; we use the normalization (1.2), for which $\mathbb{E}G = I$. Put $z_j = \sqrt{n}Q^*e_j$, so that $G = M^{-1} \sum_{j \in \mathcal{T}} z_j z_j^*$ and $\sum_{j=1}^n z_j z_j^* = nI$.

Lemma 7.1 (Deterministic Gram-error comparison). *For every sketch S ,*

$$0 \preceq R \preceq K, \quad R \preceq (I - G)K(I - G), \quad \|R\|_F^2 \leq \text{tr}(KR) \leq \text{tr}[K(I - G)K(I - G)].$$

Proof. Represent the residual by a projector. Let P be the orthogonal projector onto $\text{range}(K^{1/2}S^*)$. The identity $X(X^*X)^\dagger X^* = P_{\text{range } X}$ gives

$$\mathcal{N}_K(S^*) = K^{1/2}PK^{1/2}, \quad R = K^{1/2}(I - P)K^{1/2}.$$

Thus $0 \preceq R \preceq K$.

Insert the Gram error. Since $(I - P)K^{1/2}S^* = 0$, we have $RG = GR = 0$. Consequently

$$R = (I - G)R(I - G) \preceq (I - G)K(I - G).$$

The inequalities for the traces now follow from positivity:

$$\text{tr}(R^2) \leq \text{tr}(KR) \leq \text{tr}[K^{1/2}(I - G)K(I - G)K^{1/2}].$$

The first step uses $\text{tr}[R(K - R)] \geq 0$; the second uses the preceding Loewner comparison after congruence by $K^{1/2}$. $\square$

Lemma 7.2 (Finite-population identity). *Let $n \geq 2$, $1 \leq M \leq n$, and fix the unitary Q . For every Hermitian X ,*

$$\mathbb{E}_{\mathcal{T}}[(G - I)X(G - I)] = \lambda_{n,M} \left(\frac{1}{n} \sum_{j=1}^n z_j z_j^* X z_j z_j^* - X \right).$$

In particular $\mathbb{E}_{\mathcal{T}} \text{tr}[K(G - I)K(G - I)] = \lambda_{n,M} [n^{-1} \sum_j (z_j^ K z_j)^2 - \|K\|_F^2]$.*

Proof. For a uniform M -subset,

$$\mathbb{P}(j \in \mathcal{T}) = \frac{M}{n}, \quad \mathbb{P}(j, l \in \mathcal{T}) = \frac{M(M-1)}{n(n-1)} \quad (j \neq l).$$

Define $W = \sum_j z_j z_j^* X z_j z_j^*$. Since $\sum_j z_j z_j^* = nI$, the off-diagonal part of the double sum is

$$\sum_{j \neq l} z_j z_j^* X z_l z_l^* = n^2 X - W.$$

The two inclusion probabilities now give

$$\begin{aligned} \mathbb{E}_{\mathcal{T}} G X G &= \frac{W}{Mn} + \frac{M-1}{Mn(n-1)} (n^2 X - W) \\ &= \frac{\lambda_{n,M}}{n} W + (1 - \lambda_{n,M}) X. \end{aligned}$$

Because $\mathbb{E}_{\mathcal{T}}G = I$, expanding $(G - I)X(G - I)$ subtracts X from this expression. Taking $\text{tr}(K \cdot)$ with $X = K$ proves the scalar identity. $\square$

Lemma 7.3 (Rademacher fourth moments). *Let $\epsilon \in \{\pm 1\}^n$ be four-wise independent and unbiased. For Hermitian C , $\mathbb{E}(\epsilon^\top C \epsilon)^2 = (\text{tr } C)^2 + 2\|\Re C\|_F^2 - 2\sum_a C_{aa}^2 \leq (\text{tr } C)^2 + 2\|C\|_F^2$. For real symmetric C , $\mathbb{E}[\epsilon \epsilon^\top C \epsilon \epsilon^\top] = (\text{tr } C)I + 2C - 2\text{Diag}(C)$.*

Proof. $\mathbb{E}\epsilon_a\epsilon_b\epsilon_c\epsilon_d = \delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc} - 2\mathbf{1}_{a=b=c=d}$. Hence $\mathbb{E}(\epsilon^\top C \epsilon)^2 = (\text{tr } C)^2 + \sum_{ab}(C_{ab}^2 + |C_{ab}|^2) - 2\sum_a C_{aa}^2$, and $\sum_{ab}(C_{ab}^2 + |C_{ab}|^2) = 2\|\Re C\|_F^2$ because $C_{ba} = \overline{C_{ab}}$. The matrix identity is the same expansion with free indices. $\square$

Theorem 7.4 (One actual two-round sketch). *Let A be flat and B unitary, let D_x, D_y be four-wise independent, mutually independent and independent of $\mathcal{T}$, and let $n \geq 2$.*

$$(a) \quad \mathbb{E}\|K - \mathcal{N}_K(S^*)\|_F^2 \leq \mathbb{E}\text{tr}[K(G - I)K(G - I)] \leq \lambda_{n,M}[(\text{tr } K)^2 + \|K\|_F^2].$$

(b) *If K is real and A, B are real flat orthogonal matrices, then*

$$\mathbb{E}\text{tr}[K(G - I)K(G - I)] = \lambda_{n,M}\left[\left(1 - \frac{2}{n}\right)(\text{tr } K)^2 + \left(1 - \frac{4}{n}\right)\|K\|_F^2 + \frac{4}{n}\sum_i K_{ii}^2\right], \quad (7.1)$$

and consequently

$$\mathbb{E}\|K - \mathcal{N}_K(S^*)\|_F^2 \leq \frac{2}{M}\left(1 - \frac{M}{n}\right)(\text{tr } K)^2. \quad (7.2)$$

No independence between the sampled rows is assumed; they share both sign diagonals.

Proof. General unitary inner transform. The deterministic comparison is Lemma 7.1. To compute the remaining energy, fix a row j and condition on D_x . Define

$$V = BD_x, \quad h = \sqrt{n}A^*e_j, \quad C = D_h^*VKV^*D_h.$$

Flatness of A makes every entry of h unimodular. Therefore $z_j = V^*D_h y$, and C is unitarily similar to K . In particular,

$$z_j^*Kz_j = y^\top Cy, \quad \text{tr } C = \text{tr } K, \quad \|C\|_F = \|K\|_F.$$

The Rademacher identity of Lemma 7.3 gives

$$\mathbb{E}(z_j^*Kz_j)^2 \leq (\text{tr } K)^2 + 2\|K\|_F^2.$$

Averaging this inequality over the rows and using Lemma 7.2 proves part (a). The population identity subtracts one copy of $\|K\|_F^2$, which explains the final coefficient of that term.

Exact energy for real flat transforms. Assume now that A, B, K are real and both transforms are flat. The entries of h are signs, so multiplying y by D_h preserves its four-wise independent Rademacher law. Conditional expectation over y thus gives the equality

$$\mathbb{E}_y(z_j^\top K z_j)^2 = (\text{tr } K)^2 + 2\|K\|_F^2 - 2\sum_a (VKV^\top)_{aa}^2.$$

It remains to average the last sum over the inner signs x . For each a , write

$$(VKV^\top)_{aa} = x^\top C^{(a)}x, \quad C_{il}^{(a)} = B_{ai}K_{il}B_{al}.$$

Since $B_{ai}^2 = 1/n$, the three quantities in the fourth-moment formula are

$$\operatorname{tr} C^{(a)} = \frac{\operatorname{tr} K}{n}, \quad \|C^{(a)}\|_F^2 = \frac{\|K\|_F^2}{n^2}, \quad \sum_i (C_{ii}^{(a)})^2 = \frac{\sum_i K_{ii}^2}{n^2}.$$

Applying Lemma 7.3 a second time and summing over a yields

$$\mathbb{E}_x \sum_a (V K V^\top)_{aa}^2 = \frac{1}{n} \left[(\operatorname{tr} K)^2 + 2\|K\|_F^2 - 2 \sum_i K_{ii}^2 \right].$$

This value is independent of j . Substitute it into the preceding conditional identity, then apply Lemma 7.2; the result is (7.1).

A bound depending only on the trace. Set $T = \operatorname{tr} K$, $F = \|K\|_F^2$, and $\Delta = \sum_i K_{ii}^2$. Positivity gives $\Delta \leq F \leq T^2$. For $n \geq 4$,

$$(1 - 4/n)F + (4/n)\Delta \leq F \leq T^2.$$

For $n < 4$, the coefficient of F is negative; using $F \geq \Delta$ instead gives the same upper bound. Thus the bracket in (7.1) is at most $(2 - 2/n)T^2$. Finally,

$$\lambda_{n,M}(2 - 2/n) = \frac{2}{M}(1 - M/n),$$

which proves (7.2). □

Theorem 7.5 (The one-round law). *For the one-round sketch $S = \sqrt{n/M} R_\mathcal{T} A D_x$ with A real flat orthogonal, D_x four-wise independent and K real,*

$$\mathbb{E} \operatorname{tr}[K(G - I)K(G - I)] = \lambda_{n,M} \left[(\operatorname{tr} K)^2 + \|K\|_F^2 - 2 \sum_i K_{ii}^2 \right],$$

and (7.2) holds. Part (a) of Theorem 7.4 also covers this law (take $B = I$).

Proof. Here $z_j = D_x h$ with $h \in \{\pm 1\}^n$, and Lemma 7.3 with $C = D_h K D_h$ gives $\mathbb{E}(z_j^\top K z_j)^2 = T^2 + 2F - 2\Delta$; apply Lemma 7.2. Since $F \leq T^2$ and $\Delta \geq T^2/n$ by Cauchy–Schwarz, $T^2 + F - 2\Delta \leq (2 - \frac{2}{n})T^2$. □

Remark 7.6 (Neither law dominates; rank blindness). The one-round bracket minus the two-round bracket is $\frac{2}{n}(T^2 + 2F) - 2(1 + \frac{2}{n})\Delta$, which has either sign. For $K = e_1 e_1^\top$ the one-round energy vanishes while the two-round energy is $\lambda_{n,M}(2 - \frac{2}{n})$; for $K = \mathbf{1}\mathbf{1}^\top/n$ the two-round energy $\lambda_{n,M}(2 - \frac{6}{n} + \frac{4}{n^2})$ is the smaller one. Both satisfy the same final bound (7.2).

The quantity $\operatorname{tr}[K(G - I)K(G - I)] = \|K^{1/2} S^* S K^{1/2} - K\|_F^2$ is the approximate-matrix-multiplication error for the product $K^{1/2} \cdot K^{1/2}$ under without-replacement row sampling, and (7.2) is the corresponding variance scale $\|K^{1/2}\|_F^4/M = (\operatorname{tr} K)^2/M$ of Drineas, Kannan and Mahoney [10]. Our bound has coefficient 2 and the finite-population factor $1 - M/n$. It does not see the rank or the spectral decay of K : in the enumeration of Section 10 a rank-three K at $n = 8$, $M = 3$ has expected Nyström residual 20.0 against the bound 4250. The theorem is therefore an $O((\operatorname{tr} K)^2/M)$ statement of approximate-matrix-multiplication type, not a relative-error Nyström guarantee; its value is that it applies to one sketch with shared signs and is exact at $M = n$.

There is also a structural reason to distinguish expected and high-probability tail-relative guarantees. The finite-support obstruction in [25, proposition on expected tail-relative guarantees]

applies to the present real flat law whenever $M < n$. Choose a sketch value S_0 of positive probability and a unit vector $u \in \ker S_0$. For $K_H = H u u^\top + I_n$, the spectral tail after rank $r \geq 1$ is $n - r$, independent of H , but on the event $S = S_0$ the Nyström residual retains eigenvalue $H + 1$ along u . Its expected trace and squared Frobenius norm therefore grow without bound as $H \rightarrow \infty$. A uniform expected bound depending only on that tail is impossible. High-probability statements can instead allocate this event to their failure budget.

7.1 Trace and diagonal estimation

Corollary 7.7 (Sketch-and-correct trace estimator). *Let $\widehat{K} = \mathcal{N}_K(S^*)$ be built from M matrix–vector products with an M -row sketch satisfying (7.2), and let $\xi_1, \dots, \xi_t$ be independent Rademacher vectors, independent of S . The estimator*

$$\widehat{T} = \text{tr } \widehat{K} + \frac{1}{t} \sum_{\ell=1}^t \xi_\ell^\top (K - \widehat{K}) \xi_\ell$$

uses $M + t$ matrix–vector products, all chosen before any response is seen, and satisfies

$$\mathbb{E} \widehat{T} = \text{tr } K, \quad \text{Var } \widehat{T} \leq \frac{2}{t} \mathbb{E} \|K - \widehat{K}\|_F^2 \leq \frac{4(1 - M/n)}{Mt} (\text{tr } K)^2.$$

With $M = \min\{n, \lceil 2\sqrt{3}/\varepsilon \rceil\}$ and $t = \lceil 2\sqrt{3}/\varepsilon \rceil$ (the case $M = n$ is exact), $|\widehat{T} - \text{tr } K| \leq \varepsilon \text{tr } K$ with probability at least $2/3$.

Proof. Conditionally on S , $\widehat{T}$ is unbiased for $\text{tr } K$ and has variance $t^{-1}(2\|K - \widehat{K}\|_F^2 - 2 \sum_i (K - \widehat{K})_{ii}^2)$ by Lemma 7.3. The law of total variance and (7.2) give the bound, and Chebyshev’s inequality with $Mt \geq 12/\varepsilon^2$ gives the last claim. The quantities $\text{tr } \widehat{K}$ and $\xi^\top \widehat{K} \xi$ are computed from KS^* alone. $\square$

The $O(1/\varepsilon)$ query count is that of Hutch++ [21]; the point here is that the two-round transform realizes it with one fixed-size sketch. The Nyström approximation plus residual correction is the construction of Nyström++ [26], rather than a new estimator architecture.

Proposition 7.8 (Covariance map). *Let X be real symmetric and $1 \leq m \leq n$, and let $G_0 = S_0^\top S_0$ for an m -row sketch. For the real flat two-round law,*

$$\mathbb{E}[(G_0 - I)X(G_0 - I)] = \lambda_{n,m} \left[\left(1 - \frac{4}{n}\right)X + \left(1 - \frac{2}{n}\right)(\text{tr } X)I + \frac{4}{n} \text{Diag}(X) \right],$$

and for the real flat one-round law $\mathbb{E}[(G_0 - I)X(G_0 - I)] = \lambda_{n,m}[X + (\text{tr } X)I - 2 \text{Diag}(X)]$.

Proof. By Lemma 7.2, it suffices to compute the one-row map $\mathbb{E}[z_j z_j^\top X z_j z_j^\top]$. The finite-population correction then subtracts X and multiplies by $\lambda_{n,m}$.

For the two-round law, condition on x and set $V = BD_x$. Then $z_j = V^\top \epsilon$ for a four-wise independent sign vector ϵ . The matrix fourth-moment identity gives

$$\mathbb{E}_y[z_j z_j^\top X z_j z_j^\top] = (\text{tr } X)I + 2X - 2V^\top \text{Diag}(V X V^\top) V.$$

The (i, l) entry of the last term before its factor -2 is

$$x_i x_l \sum_a B_{ai} B_{al} \sum_{u,v} B_{au} x_u X_{uv} x_v B_{av}.$$

Apply the four-point sign formula to $x_i x_l x_u x_v$, using $B_{ai}^2 = 1/n$. This gives

$$\mathbb{E}_x[V^\top \text{Diag}(V X V^\top) V]_{il} = \frac{1}{n} [(\text{tr } X) \delta_{il} + 2X_{il} - 2\delta_{il} X_{ii}].$$

Consequently the unconditional one-row map is

$$(1 - 2/n)[(\text{tr } X)I + 2X] + (4/n) \text{Diag}(X).$$

Subtracting X proves the two-round formula.

For the one-round law, $z_j = D_h x$ with h a fixed sign vector. The fourth-moment identity directly gives the one-row map $(\text{tr } X)I + 2X - 2\text{Diag}(X)$. Again subtract X and apply the finite-population factor. $\square$

Corollary 7.9 (Sketch-and-correct diagonal estimator). *Let K be real, $\widehat{K}$ as in Corollary 7.7, and let S_0 be an independent m -row sketch of either real flat law. Then $\widehat{d} = \text{diag}(\widehat{K}) + \text{diag}((K - \widehat{K})S_0^\top S_0)$ uses $M + m$ matrix-vector products, is unbiased for $\text{diag } K$, and*

$$\mathbb{E}\|\widehat{d} - \text{diag } K\|_2^2 \leq \frac{2}{m} \left(1 - \frac{m}{n}\right) \mathbb{E}\|K - \widehat{K}\|_F^2 \leq \frac{4}{mM} \left(1 - \frac{m}{n}\right) \left(1 - \frac{M}{n}\right) (\text{tr } K)^2.$$

Proof. Condition on $\widehat{K}$ and put $X = K - \widehat{K}$, $r_i = X e_i$. The error in coordinate i is $r_i^\top (G_0 - I) e_i$, whose second moment is $e_i^\top \mathbb{E}[(G_0 - I) r_i r_i^\top (G_0 - I)] e_i$. By Proposition 7.8 this equals $\lambda_{n,m}[X_{ii}^2 + (1 - \frac{2}{n})\|r_i\|^2]$ for the two-round law and $\lambda_{n,m}[\|r_i\|^2 - X_{ii}^2]$ for the one-round law. Summing over i , the total is at most $\lambda_{n,m}(2 - \frac{2}{n})\|X\|_F^2 = \frac{2}{m}(1 - \frac{m}{n})\|X\|_F^2$. Now apply (7.2). $\square$

With $M = m = \min\{n, \lceil 2\sqrt{3}/\varepsilon \rceil\}$, using exact recovery if this equals n , the bound gives $\|\widehat{d} - \text{diag } K\|_2 \leq \varepsilon \text{tr } K$ with probability at least $2/3$, the guarantee of $\text{Diag}++$ [4]. The two-round law has a nonzero error even for diagonal X (the term X_{ii}^2), which the one-round law does not.

8 Fresh-row drift

Lemma 8.1 (One Khintchine factor). *Let A be flat, B unitary, the signs fully independent, j fixed and $z = \sqrt{n} Q^* e_j$. Then $\mathbb{E} z z^* = I$, and for $p \geq 1$, $u \in \mathbb{C}^n$, $X \succeq 0$,*

$$\mathbb{E}|u^* z|^{2p} \leq g(p) \|u\|^{2p}, \quad \mathbb{E}(z^* X z)^p \leq g(p) (\text{tr } X)^p, \quad g(p) = \frac{2^p \Gamma(p + \frac{1}{2})}{\sqrt{\pi}}.$$

Proof. Condition on D_x and put $v = B D_x u$. Then $\overline{u^* z} = \sqrt{n} (Q u)_j = \sum_a y_a c_a$ with $c_a = \sqrt{n} A_{ja} v_a$, and $\sum_a |c_a|^2 = \|v\|^2 = \|u\|^2$ by flatness. The same computation gives $\mathbb{E}_y z z^* = I$. **Linear moment bound.** For real coefficients, the sharp Khintchine inequality of Haagerup [13] (for exponents in $(2, 3)$ see also Mordhorst [22]) gives $\mathbb{E} |\sum_a y_a c_a|^{2p} \leq \mathbb{E} |g|^{2p} \|c\|^{2p}$ with g standard Gaussian, and $\mathbb{E} |g|^{2p} = g(p)$. For complex coefficients apply it to the real and imaginary parts and Minkowski's inequality in L^p to $|\sum y_a c_a|^2 = (\sum y_a \Re c_a)^2 + (\sum y_a \Im c_a)^2$. **Quadratic moment bound.** Write $X = \sum_i \mu_i u_i u_i^*$ with orthonormal u_i and $\mu_i \geq 0$. Minkowski's inequality and the linear bound give

$$\|(z^* X z)\|_{L^p} \leq \sum_i \mu_i \| |u_i^* z|^2 \|_{L^p} \leq g(p)^{1/p} \sum_i \mu_i.$$

Taking the p th power gives the stated bound. $\square$

Since $g(1) = 1$ and $\frac{d}{dp} \log g(p)|_{p=1} = \log 2 + \psi(3/2) = 2 - \gamma - \log 2$,

$$\kappa_0 := \lim_{\beta \downarrow 0} g(1 + \beta)^{1/\beta} = \frac{e^{2-\gamma}}{2} = 2.0743278106\dots, \quad (8.1)$$

where γ is Euler's constant.

Lemma 8.2 (Ratio conversion; the argument of [19]). *Let z be isotropic with $\mathbb{E}|u^*z|^{2p} \leq H(p)\|u\|^{2p}$ and $\mathbb{E}(z^*Xz)^p \leq H(p)(\text{tr } X)^p$ at $p = 1 + \beta$, $\beta > 0$. For nonzero PSD Y, R with $\ker R \subseteq \ker Y$,*

$$\mathbb{E} \frac{z^*Yz}{z^*Rz} \geq \frac{\text{tr } Y}{H(1 + \beta)^{1/\beta} \text{tr } R}$$

(the quotient being 0 when $z^*Rz = 0$).

Proof. Set $a = z^*Yz$ and $b = z^*Rz$. The kernel condition ensures $a = 0$ whenever $b = 0$, while isotropy gives $\mathbb{E}a = \text{tr } Y$. Introduce the probability measure

$$dQ = \frac{a}{\text{tr } Y} d\mathbb{P}.$$

Under Q , the variable b is positive almost surely. Jensen's inequality for $t \mapsto t^{-1/\beta}$ therefore gives

$$\mathbb{E} \frac{a}{b} = \text{tr } Y \mathbb{E}_Q b^{-1} \geq \text{tr } Y (\mathbb{E}_Q b^\beta)^{-1/\beta}.$$

It remains to bound the last expectation. Write $Y = \sum_i \mu_i u_i u_i^*$ with orthonormal u_i and $\mu_i \geq 0$. Hölder's inequality with exponents $p = 1 + \beta$ and p/β gives

$$\begin{aligned} \mathbb{E}[|u_i^*z|^2 b^\beta] &\leq (\mathbb{E}|u_i^*z|^{2p})^{1/p} (\mathbb{E} b^p)^{\beta/p} \\ &\leq H(p)(\text{tr } R)^\beta. \end{aligned}$$

Multiply by μ_i , sum over i , and divide by $\text{tr } Y$ to obtain $\mathbb{E}_Q b^\beta \leq H(p)(\text{tr } R)^\beta$. Substitution proves the claim. $\square$

Theorem 8.3 (Fresh-row drift). *Let z be as in Lemma 8.1, $R_0 \succeq 0$ nonzero and $R_1 = R_0 - R_0 z(z^* R_0 z)^\dagger z^* R_0$. Then $0 \preceq R_1 \preceq R_0$ and*

$$\mathbb{E}(\text{tr } R_0 - \text{tr } R_1) \geq \frac{\text{tr}(R_0^2)}{\kappa_0 \text{tr } R_0}.$$

If R_k is obtained from $K \succeq 0$ by k such updates with independent fresh draws of (D_x, D_y) , then $\mathbb{E}\|R_k\|_F^2 \leq \kappa_0(\text{tr } K)^2/k$.

Proof. The trace drop is $z^* R_0^2 z / z^* R_0 z$. Apply Lemma 8.2 with $Y = R_0^2$, $R = R_0$, $H = g$, and let $\beta \downarrow 0$ using (8.1). For the second claim, conditioning gives $\mathbb{E}[\text{tr } R_t - \text{tr } R_{t+1} \mid R_t] \geq \text{tr}(R_t^2)/(\kappa_0 \text{tr } K)$; summing, $\sum_{t < k} \mathbb{E} \text{tr}(R_t^2) \leq \kappa_0(\text{tr } K)^2$, and $\text{tr}(R_t^2)$ is nonincreasing because $R_{t+1} \preceq R_t$. $\square$

Remark 8.4 (What the constant means). The constant κ_0 is exactly the constant of an i.i.d. Rademacher probe, which satisfies the same Khintchine envelope, costs $O(n)$ and needs no flatness. Theorem 8.3 therefore says that a fresh row of the two-round transform is no worse than a Rademacher probe, and in particular that the two transforms do not square the constant. It is not an algorithmic gain over Rademacher probing; the SRHT-specific content is in Theorem 7.4.

The theorem does not justify charging the M rows of one sketch as M independent updates, since they share D_x, D_y . Finite spherical constants do not transfer either: at $n = 2$, $H_2 D_y H_2$ is a signed identity or a signed swap, so $|e_1^* z|^2 \in \{0, 2\}$ equiprobably and the order-one constant of this law is 2, above the value $e/2$ of the two-dimensional real sphere. For a nonflat outer transform and a uniformly random row J , the same proof with $H(p) = g(p)\alpha^{p-1}$ gives the constant $\alpha\kappa_0$.

9 Sketched randomly pivoted Cholesky

Let $F \in \mathbb{R}^{D \times N}$ and $A = F^\top F$. *Sketched RP* runs RP on $B = (\Pi F)^\top (\Pi F)$ for a random sketch Π with D columns and measures the error on A . In *RPCholesky under approximate and sketched pivot probabilities* [18] it is shown that one does not need an embedding of range F : it suffices that Π embeds, with a failure probability small against the RP path likelihood, the spans of the pivot columns actually reached, augmented by one direction. This section records what the two-round transform supplies to that framework.

Definition 9.1 (Path-local moment property). A random $\Pi \in \mathbb{C}^{M \times D}$ has the moment property $(\ell, q, \varepsilon_0, C_0)$ if $\mathbb{E} \operatorname{tr}(U^* \Pi^* \Pi U - I_d)^{2q} \leq C_0 d \varepsilon_0^{2q}$ for every fixed isometry $U \in \mathbb{C}^{D \times d}$ with $d \leq \ell$.

Proposition 9.2 (The two-round transform). Let $0 < \varepsilon_0 < 1/2$, $\ell \geq 1$, $q \geq 1$, and let $\Lambda_q(\ell)$ denote Λ_q with $d = \ell$. The two-round sketch on $\mathbb{C}^D$ with $M \geq \min\{D, \lceil 5\Lambda_q(\ell)/\varepsilon_0^2 \rceil\}$ rows has the moment property $(\ell, q, \varepsilon_0, 6/5)$, under $\min(D, 4q)$ -wise independent signs.

Proof. Λ_q is increasing in d , and $5\Lambda_q(\ell)/\varepsilon_0^2 = 20\Lambda_q(\ell)/(2\varepsilon_0)^2$. Apply (4.1) with $\varepsilon = 2\varepsilon_0 < 1$ to each $d \leq \ell$. $\square$

Imported theorem 9.3 (Sketched RP, conservative path interface; [18]). Let $1 \leq r < k$, $\tau = \tau_r(A) > 0$, $0 < \varepsilon \leq 1$ and $0 < \eta \leq 1/4$, and write $R_A(J) = A - A_{:,J} A_{J,J}^\dagger A_{J,:}$. If Π is independent of the pivoting randomness and has the moment property $(k+1, q, \varepsilon_0, C_0)$ with $q \geq \max\{2, k\}$ and $\varepsilon_0 \leq \eta/2$, then on a good event for the current pivot set J the sketched residual trace satisfies $(1 - \eta - \eta^2) \operatorname{tr} R_A(J) \leq \operatorname{tr} R_B(J) \leq (1 + \eta) \operatorname{tr} R_A(J)$, and the expected decrease of $\operatorname{tr} R_A$ in one sketched step is at least $\gamma \operatorname{tr}(R_A^2)/\operatorname{tr} R_A$ with $\gamma = (1 - \eta - \eta^2)/(1 + \eta)$. Hence the recurrence argument for randomly pivoted Cholesky [7], run with the factor γ as in [18], gives $\mathbb{E} \operatorname{tr} R_A(J_k) \leq (1 + \varepsilon)\tau + \operatorname{tr}(A) P_{\text{fail}}$, for instance for $k \geq \lceil (r/\gamma)(\varepsilon^{-1} + \log \varepsilon^{-1} + \log(\operatorname{tr} A/\tau)) \rceil$. The path second-moment estimate gives the explicit failure bound

$$P_{\text{fail}} \leq \min\left\{1, 2(k+1)e^{k\chi_\eta^2/2} C_0^{1/2} 2^{-q/2}\right\}, \quad \chi_\eta = \frac{2\eta}{1-\eta}.$$

The hypotheses and the exhausted-residual convention are given in [18]. The same sketch can be reused for the reconstruction $F \approx F_J(\Pi F_J)^\dagger \Pi F$.

Remark 9.4 (Rows scale quadratically in the pivot budget). For $\varepsilon_0 = \eta/2$ and $C_0 = 6/5$, the explicit path bound makes $\operatorname{tr}(A)P_{\text{fail}} \leq \varepsilon\tau$ whenever

$$q \geq \max\left\{2, k, \frac{k\chi_\eta^2}{\log 2} + 2 \log_2 \frac{2(k+1)\sqrt{6/5} \operatorname{tr} A}{\varepsilon\tau}\right\}.$$

The constraint $q \geq k$ covers the determinant path estimate; it remains in this conservative interface even when the likelihood exponent is small. Thus a sufficient order is $O(k + \log(k \operatorname{tr} A/(\varepsilon\tau)))$. For the two-round transform with flat factors, $\Lambda_q(k+1) \geq 2(k+1) + 22q^2 - q$, so Proposition 9.2

gives $M = O(\eta^{-2}[k + (k + \log(k \operatorname{tr} A/(\varepsilon\tau)))^2])$ rows: quadratic in k at fixed η , whereas a sketch with allowance linear in q gives rows linear in k . The uncapped sufficient budget is independent of N and D . The hybrid path comparison in [18] can instead use $q \geq 2$ and order $O(\eta^2 k + \log(k \operatorname{tr} A/(\varepsilon\tau)))$; its SRHT budget is still quadratic in k at fixed η . Numerically, at $\eta = 1/4$ ($\varepsilon_0 = 1/8$) and $q = \ell$, Proposition 9.2 asks for $M/\ell^2 \approx 1.1 \times 10^4$ for $\ell = 11, 51, 101$ (Section 10); this is a scope theorem, useful only for very tall F .

10 Numerical evaluation

The original tables were computed in Julia with 256-bit floating point; the scripts are supplied in `checks/original/`. An independent Python audit in `checks/audit_srht.py` recomputes the scalar prescriptions, certificate boundaries, and representative whole-sketch identities. These are numerical checks of the stated formulas, not a formal verification of the imported operator results.

In the original computation, binomial atoms were evaluated from exact log-factorials (Stirling’s series with remainder below 10^{-30} for arguments above 10^3). They evaluate the sufficient conditions proved above and are not empirical thresholds. Unless stated otherwise the transforms are flat, $n = 2^{30}$ (no prescription is capped), $\varepsilon = 1/2$ and $\delta = 0.01$; then $q = \lceil \log_4(400d) \rceil$ is the order of [15, 14], $q_* = \lceil \log_4(120d) \rceil$ and $q_f = \lceil \log_4(100d) \rceil$.

d	q/q_*	$216D_q$	$72\Lambda_q$	$60D_{q_*}$	$20\Lambda_{q_*}$	direct M	q (indep.)	least feasible
10	6/6	381 888	369 386	106 080	102 608	58 406	4 (16)	58 396
100	8/7	749 952	717 818	165 120	157 664	125 760	6 (24)	125 750
1 000	10/9	1 900 800	1 733 312	473 280	427 412	346 325	10 (40)	346 315
10 000	11/11	9 894 528	7 894 646	2 748 480	2 192 958	1 423 405	18 (72)	1 423 395

Table 1: Sufficient row counts. The first two columns after q/q_* calibrate the crude root at the order q of [15, 14]; the next two the exact root at q_* . “Direct” is the output of Algorithm 1 with safety margin $s = 10$, its order and the required sign independence $\min(n, 4q)$; the certificates at the returned M are 0.0099926, 0.0099947, 0.0099969, 0.0099986. “Least feasible” is the smallest M with $\operatorname{Cert}(q, M) \leq \delta$ for some $q \leq 40$, confirmed in the original diagnostic run by an exhaustive scan of all M below the bisection value (certificates at least feasible minus one: 0.0100006, 0.0100002, 0.0100003, 0.0100001).

For the canonical prescription the exact certificates at $M = \lceil 20\Lambda_{q_*}/\varepsilon^2 \rceil$ are $1.2 \cdot 10^{-3}$, $2.8 \cdot 10^{-3}$, $1.4 \cdot 10^{-3}$ and $6.6 \cdot 10^{-4}$ for the four values of d , well below $\delta = 0.01$; this is the slack recovered by the selector. The selected orders are larger than q_* for large d : at $d = 10^4$ the selector trades the order $11 \rightarrow 18$ for 35% fewer rows.

Whole-sketch identities (numerical verification). We enumerated the complete real flat Walsh laws: at $n = 4$ all $2^4 \cdot 2^4$ sign pairs and all M -subsets in exact rational arithmetic, and at $n = 8$ in double precision, for integer PSD test matrices. The identities (7.1), Theorem 7.5 and Proposition 7.8 held exactly at $n = 4$ for $M = 1, 2, 3$ and to 10^{-9} at $n = 8$. Table 5 shows representative values; the Nyström residual was below the Gram-error energy and below (7.2) in every case.

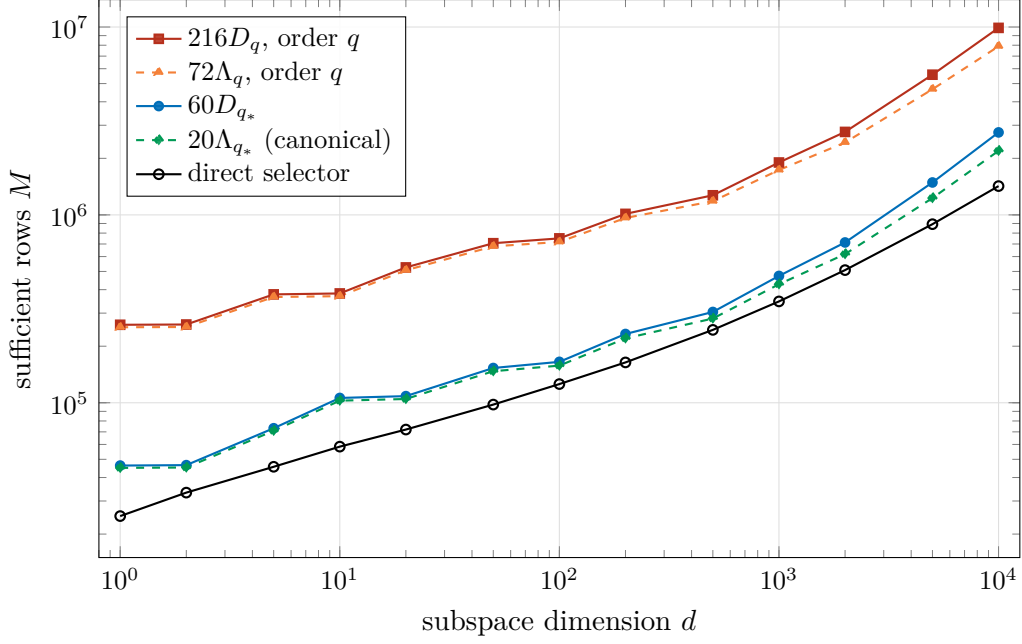


Figure 3: Sufficient rows against d ($n = 2^{30}$, $\varepsilon = 1/2$, $\delta = 0.01$, flat transforms). The steps of the upper curves are jumps of the moment order. The selector orders are 2, 3, 3, 4, 4, 5, 6, 6, 8, 10, 12, 15, 18 at the plotted d .

d	q	q_f	$72\Lambda_q$	$72\Lambda_{q_f}$	saving	$20\Lambda_{q*}$
1	5	4	252 288	162 041	35.8%	45 012
10	6	5	369 386	259 190	29.8%	102 608
100	8	7	717 818	567 591	20.9%	157 664
1 000	10	9	1 733 312	1 538 681	11.2%	427 412
10 000	11	10	7 894 646	7 628 223	3.4%	2 192 958

Table 2: The reduced order of Corollary 4.4. Dropping one order in the crude-root prescription saves up to about a third of the rows at small d ; the canonical prescription is smaller still.

Scalar checks. $2/\sqrt{20} + 1/20 = 0.49721$, $\sqrt{2}/3 + 1/36 = 0.49918$, $12/\sqrt{646} + 18/646 = 0.4999975$, $\sqrt{\pi}/24 = 0.07385 < 1/12$. The atom bound of Lemma 3.2 was checked for all $2 \leq n \leq 400$, $1 \leq M < n$ (maximal ratio of atom to bound 1, attained at $n = 2$, $M = 1$ where both equal $1/2$). On a grid of coherences $\alpha \in [1, 10]$, $\beta \in [1, 5]$, $q \leq 30$, the ratio Λ_q/D_q ranged over $[2.07, 3.00]$. Numerically $g(1 + \beta)^{1/\beta} = 2.08402, 2.07442, 2.0743278$ at $\beta = 10^{-2}, 10^{-4}, 10^{-8}$. For sketched RP (Proposition 9.2) at $\varepsilon_0 = 1/8$ and $q = \ell$ the row counts are $1.36 \cdot 10^6$, $2.89 \cdot 10^7$ and $1.13 \cdot 10^8$ for $\ell = 11, 51, 101$.

Confidence price	$\bar{\Gamma}$	q_*	$20\Lambda_{q_*}$	$60D_{q_*}$	direct (q)
fixed frame	1	7	157 664	165 120	125 760 (6)
two flat bands, 5 + 5	252	11	357 798	372 480	276 534 (8)
arbitrary spectrum	10!	18	921 368	957 120	696 744 (12)
dictionary union bound	$\binom{1000}{10} \approx 2.6 \cdot 10^{23}$	46	5 892 073	6 118 080	4 467 916 (31)

Table 3: RP-selected frames (Corollary 6.2) for $d = 100$, $k = 10$, $N = 1000$. Without any spectral information the factorial gate uses 84% fewer rows than the union bound over the dictionary. The direct column runs Algorithm 1 at confidence $\delta/\bar{\Gamma}$.

d	q_* at $\delta/2$	$20\Lambda_{q_*}$	q_* at $8\delta^2/9$	$20\Lambda_{q_*}$
10	6	102 608	9	227 705
100	8	199 394	11	357 798
1 000	9	427 412	12	605 597
10 000	11	2 192 958	14	2 442 162

Table 4: Rank-two selection (Proposition 6.3): the pointwise allowance $\delta/2$ against the second-moment allowance $8\delta^2/9$.

case	law	Gram energy	formula	$\mathbb{E}\ K - \mathcal{N}_K\ _F^2$	bound (7.2)
$n = 4$, $M = 1$, rank 3	two-round	2203.5	2203.5	556.8	4213.5
	one-round	2738.0	2738.0	717.2	4213.5
$n = 4$, $M = 2$, rank 3	two-round	734.5	734.5	167.1	1404.5
	one-round	912.67	912.67	280.9	1404.5
$n = 8$, $M = 3$, rank 3	two-round	2512.32	2512.32	20.0	4250.4
	one-round	2620.95	2620.95	2.5	4250.4
$n = 8$, $M = 3$, $K = \mathbf{1}\mathbf{1}^\top/n$	two-round	0.3125	0.3125	0.038	0.4167
	one-round	0.4167	0.4167	0.070	0.4167
$n = 8$, $M = 3$, $K = e_1 e_1^\top$	two-round	0.4167	0.4167	0	0.4167
	one-round	0	0	0	0.4167

Table 5: Exact enumeration of the whole-sketch law (numerical). “Gram energy” is $\mathbb{E} \text{tr}[K(G - I)K(G - I)]$ and “formula” is (7.1) or Theorem 7.5. The last two cases show that neither Gram-error energy dominates the other (Remark 7.6); the rank-three rows show how far the rank-blind bound is from the true Nyström residual.

11 Open problems

1. *The confidence term.* All prescriptions keep the allowance $d + 12q^2$ with $q \asymp \log(d/\delta)$. The support-alignment example of [16, 15] rules out a uniform $d + \log(1/\delta)$ term for any fixed number of transform-sign layers, but the gap between $\log^2$ and the obstruction is open, as is the constant 12.
2. *Efficient search of the certified widths.* The uncapped root is globally nonmonotone, as Remark 5.2 shows. Determine whether its feasible set has a monotone structure in the useful regime, or use the stronger imported allowance $L_q \leq \min\{1, \Lambda_q/n\}$ to obtain a search guarantee. A proof is needed before bisection can be asserted to return a global minimum.
3. *A tail-relative high-probability whole-sketch bound.* Theorem 7.4 is rank-blind, and the finite-support obstruction of Remark 7.6 rules out a uniform expected bound depending only on a spectral tail when $M < n$. Determine a high-probability relative trace or Frobenius bound for one two-round sketch with shared signs, with explicit dependence on rank, distortion and failure probability.
4. *Rows linear in the pivot budget.* The hybrid path comparison in [18] already reduces the order to $O(\eta^2 k + \log(k \operatorname{tr} A/(\varepsilon\tau)))$, but the certified two-round SRHT budget remains quadratic in k at fixed distortion. Can a sharper embedding or path comparison yield a budget linear in k ?
5. *Beyond the fixed-range gate.* The gate needs full rank, a fixed range and an unordered release. A price for partial-rank or moving-range selection would cover the natural pipeline in which the frame is the span of the first pivots.
6. *Block drift.* The whole-block mixed moment of the real flat law suggests a drift constant $1 + \frac{2}{M}(1 - \frac{M}{n})$ for a full block of M shared rows. A proof with explicit dependence on the shared signs would connect Theorem 8.3 and Theorem 7.4.

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